MODEL LAB · 2026-10-26

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

19,871
held-out plate appearances

2.97%
held-out HR rate

0.5993
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-11, scored on the 19,871 that came after (actual HR rate 2.97%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5817	1.49×	0.02881	0.13302	0.00516
KNN comps Prior is the pooled rate of the nearest comparables.	0.5993	1.56×	0.02876	0.13194	0.00482
SVR Support-vector regression on the same feature matrix.	0.5936	1.47×	0.02880	0.13266	0.00607
Random forest Forest regression; a learned similarity metric.	0.5970	1.51×	0.02877	0.13209	0.00471
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5993	1.56×	0.02876	0.13194	0.00482
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5993	1.56×	0.02876	0.13194	0.00482
Linear regression Ordinary least squares on the same features.	0.5964	1.52×	0.02877	0.13227	0.00517
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5963	1.57×	0.02879	0.13238	0.00554
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5960	1.52×	0.02878	0.13224	0.00546
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5947	1.49×	0.02879	0.13246	0.00548

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **257** knn **257** svr **257** rf **257** core **257** ensemble **257** linear **257** logistic **257** nn **257** xgb **257**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (1,863 PA, actual HR rate 2.47%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.6012	1.31×	0.02408	0.11577	0.01650
KNN comps Prior is the pooled rate of the nearest comparables.	0.6428	1.96×	0.02394	0.11263	0.00990
SVR Support-vector regression on the same feature matrix.	0.6103	1.74×	0.02415	0.11730	0.01734
Random forest Forest regression; a learned similarity metric.	0.6393	1.96×	0.02397	0.11299	0.01039
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6428	1.96×	0.02394	0.11263	0.00990
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6428	1.96×	0.02394	0.11263	0.00990
Linear regression Ordinary least squares on the same features.	0.6419	1.96×	0.02400	0.11412	0.01415
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6348	1.96×	0.02404	0.11408	0.01429
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6323	1.96×	0.02397	0.11350	0.01510
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6282	1.74×	0.02409	0.11527	0.01516

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Jackson Chourio 16.9%, Junior Caminero 16.1%, William Contreras 15.3%, Jake Bauers 15.0%, Ryan Vilade 14.5%, Yandy Díaz 13.1%
Flat league	6/6	Jackson Chourio 17.1%, Junior Caminero 15.2%, William Contreras 15.1%, Jake Bauers 14.0%, Ryan Vilade 13.8%, Yandy Díaz 13.5%
KNN comps	6/6	Jackson Chourio 17.2%, Junior Caminero 15.9%, William Contreras 15.2%, Jake Bauers 15.2%, Ryan Vilade 14.4%, Yandy Díaz 13.2%
SVR	6/6	Jackson Chourio 17.8%, Junior Caminero 17.5%, William Contreras 15.2%, Jake Bauers 15.0%, Ryan Vilade 14.1%, Yandy Díaz 13.8%
Random forest	6/6	Jackson Chourio 17.7%, Junior Caminero 16.4%, Jake Bauers 15.2%, William Contreras 15.1%, Ryan Vilade 14.6%, Yandy Díaz 13.5%
Core (KNN+SVR)	6/6	Jackson Chourio 17.2%, Junior Caminero 15.9%, William Contreras 15.2%, Jake Bauers 15.2%, Ryan Vilade 14.4%, Yandy Díaz 13.2%
Forest blend	6/6	Jackson Chourio 17.2%, Junior Caminero 15.9%, William Contreras 15.2%, Jake Bauers 15.2%, Ryan Vilade 14.4%, Yandy Díaz 13.2%
Linear regression	6/6	Jackson Chourio 17.5%, Junior Caminero 16.1%, William Contreras 15.2%, Jake Bauers 14.9%, Ryan Vilade 14.1%, Yandy Díaz 13.4%
Logistic regression	6/6	Jackson Chourio 17.6%, Junior Caminero 16.7%, William Contreras 15.2%, Jake Bauers 15.0%, Ryan Vilade 13.9%, Yandy Díaz 13.2%
Neural net (torch)	6/6	Jackson Chourio 17.6%, Junior Caminero 16.6%, William Contreras 15.3%, Jake Bauers 15.1%, Ryan Vilade 14.3%, Yandy Díaz 13.2%
XGBoost	6/6	Jackson Chourio 17.7%, Junior Caminero 17.0%, Jake Bauers 15.2%, William Contreras 15.0%, Ryan Vilade 14.4%, Yandy Díaz 13.8%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	**Jackson Chourio** MIL Slate	**16.9%**	17.1%	17.2%	17.8%	17.7%	17.2%	17.2%	17.5%	17.6%	
2	**Junior Caminero** TB Slate	**16.1%**	15.2%	15.9%	17.5%	16.4%	15.9%	15.9%	16.1%	16.7%	
3	**William Contreras** MIL	**15.3%**	15.1%	15.2%	15.2%	15.1%	15.2%	15.2%	15.2%	15.2%	
4	**Jake Bauers** MIL	**15.0%**	14.0%	15.2%	15.0%	15.2%	15.2%	15.2%	14.9%	15.0%	
5	**Ryan Vilade** TB	**14.5%**	13.8%	14.4%	14.1%	14.6%	14.4%	14.4%	14.1%	13.9%	
6	**Yandy Díaz** TB	**13.1%**	13.5%	13.2%	13.8%	13.5%	13.2%	13.2%	13.4%	13.2%	
7	**Christian Yelich** MIL	**12.4%**	12.5%	12.8%	12.1%	12.5%	12.8%	12.8%	12.7%	12.1%	
8	**Garrett Mitchell** MIL	**12.0%**	11.5%	11.8%	10.8%	11.4%	11.8%	11.8%	11.5%	11.4%	
9	**Brice Turang** MIL	**11.7%**	11.9%	11.4%	11.5%	11.3%	11.4%	11.4%	11.0%	10.9%	
10	**Victor Mesa Jr.** TB	**9.7%**	9.4%	9.7%	10.1%	10.1%	9.7%	9.7%	9.9%	9.8%	
11	**Nick Fortes** TB	**8.4%**	9.7%	8.4%	8.3%	8.1%	8.4%	8.4%	9.2%	9.0%	
12	**Jonny DeLuca** TB	**8.2%**	9.0%	7.8%	7.9%	7.5%	7.8%	7.8%	7.9%	8.2%	
13	**Cooper Pratt** MIL	**7.0%**	9.0%	7.0%	6.3%	6.7%	7.0%	7.0%	6.1%	6.3%	
14	**David Hamilton** MIL	**6.6%**	7.9%	6.5%	6.1%	5.8%	6.5%	6.5%	6.3%	6.2%	
15	**Sal Frelick** MIL	**6.5%**	8.1%	6.7%	6.1%	6.1%	6.7%	6.7%	6.5%	6.6%	
16	**Ben Williamson** TB	**6.3%**	9.9%	6.8%	6.2%	5.9%	6.8%	6.8%	4.5%	6.6%	
17	**Taylor Walls** TB	**5.6%**	7.1%	5.4%	4.3%	4.4%	5.4%	5.4%	5.4%	5.3%	
18	**Chandler Simpson** TB	**2.7%**	6.0%	3.6%	1.5%	1.6%	3.6%	3.6%	1.5%	3.2%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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